SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA06	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	**Assessment:**	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	**Total Marks:**	100 (2 Questions × 50 Marks)
CO1 Statement:	*Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)*		
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Section / Batch:	2025	**Date:**	22/07/2026

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
	GRAND TOTAL:	**100 Marks**	

SECTION A – APPLICATION BASED QUESTIONS

Q77. Application: E-Library Search System
An e-library reduces the collection size by half after each comparison while searching for a title. Define divide-and-conquer strategy, explain how repeated halving reduces the problem size, and calculate the number of
steps. Determine both the time complexity and space complexity of this search system.

Q78. Application: Distributed Data Processing System
A distributed system processes data blocks using the recurrence $T(n) = T(n/2) + n$. Recall recurrence solving
methods, explain how work grows across recursive calls, and determine the asymptotic time complexity. Also
analyze the space complexity of the algorithm.

Code of Conduct

I M Vishnu Vardhan- 192472087 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student M Vishnu

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (50 Marks)						
Q2 (50 Marks)						
					Overall Total (100)	

Faculty In-charge	Course Coordinator	Head of Department
Signature: ________________	Signature: ________________	Signature: ________________
Date: ________________	Date: ________________	Date: ________________

Q77. Application: E-Library Search System

Question

An e-library reduces the collection size by half after each comparison while searching for a title. Define divide-and-conquer strategy, explain how repeated halving reduces the problem size, and calculate the number of steps. Determine both the time complexity and space complexity of this search system.

Answer

The searching technique described is **Binary Search**, which follows the **Divide-and-Conquer** strategy.

1. Divide-and-Conquer Strategy

Divide-and-conquer is an algorithm design technique in which a problem is:

1. **Divided** into smaller subproblems.
2. **Conquered** by solving the smaller subproblem recursively.
3. **Combined** to obtain the final solution.

In binary search, the sorted collection is divided into two equal halves after every comparison. Only the half that may contain the required title is searched further.

2. Repeated Halving

Suppose the e-library contains **n** titles.

After each comparison:

- Initially: n
- After 1 comparison: n/2
- After 2 comparisons: n/4
- After 3 comparisons: n/8
- After k comparisons: n/2^k

The search ends when only one element remains:

$$\frac{n}{2^k} = 1$$

Therefore,

$$n = 2^k$$

Taking logarithm on both sides:

$$k = \log_2 n$$

Hence, the number of comparisons or steps is approximately:

$$\boxed{\log_2 n}$$

3. Example

If there are **1024 titles**:

$$1024 \to 512 \to 256 \to 128 \to 64$$
$$\to 32 \to 16 \to 8 \to 4 \to 2 \to 1$$

Number of steps:

$$\log_2 1024 = 10$$

Thus, at most **10 comparisons** are needed.

4. Time Complexity

The recurrence relation is:

$$T(n) = T(n/2) + O(1)$$

Using the recurrence expansion:

$$T(n) = T(n/4) + 2$$
$$T(n) = T(n/8) + 3$$

After k steps:

$$T(n) = T(n/2^k) + k$$

When:

$$n/2^k = 1$$

we get:

$$k = \log_2 n$$

Therefore:

$$\boxed{T(n) = O(\log n)}$$

- **Best case:** $O(1)$, if the title is found in the first comparison.
- **Average case:** $O(\log n)$
- **Worst case:** $O(\log n)$

5. Space Complexity

For **iterative binary search**, only a constant amount of extra memory is used.

$$\boxed{S(n) = O(1)}$$

For **recursive binary search**, the recursive calls form a stack of depth $\log n$.

$$\boxed{S(n) = O(\log n)}$$

Final Answer

Aspect	**Complexity**
Searching technique	Binary Search
Strategy	Divide and Conquer
Number of steps	$\log_2 n$
Best-case time	$O(1)$

Aspect	Complexity
Average-case time	$O(\log n)$
Worst-case time	$O(\log n)$
Space (Iterative)	$O(1)$
Space (Recursive)	$O(\log n)$

Conclusion: The e-library search system uses a divide-and-conquer approach by repeatedly reducing the search space to half. Hence, the search requires $O(\log n)$**time** and $O(1)$**auxiliary space for an iterative implementation**.

Q78. Application: Distributed Data Processing System

Question

A distributed system processes data blocks using the recurrence:

$$T(n) = T(n/2) + n$$

Recall recurrence solving methods, explain how work grows across recursive calls, and determine the asymptotic time complexity. Also analyze the space complexity of the algorithm.

Answer

The given recurrence is:

$$T(n) = T(n/2) + n$$

This means that the algorithm processes a problem of size **n**, recursively processes a problem of size **n/2**, and performs **n units of additional work** at the current level.

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1. Recurrence Expansion Method

Given:

$$T(n) = T(n/2) + n$$

Expand the recurrence:

$$T(n) = T(n/4) + \frac{n}{2} + n$$

Again:

$$T(n) = T(n/8) + \frac{n}{4} + \frac{n}{2} + n$$

Continuing:

$$T(n) = T(n/2^k) + n\left(1 + \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^{k-1}}\right)$$

The recursion ends when:

$$\frac{n}{2^k} = 1$$

Therefore:

$$k = \log_2 n$$

The total work is:

$$T(n) = n\left(1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots\right)$$

This is a geometric series.

$$1 + \frac{1}{2} + \frac{1}{4} + \cdots < 2$$

Therefore:

$$T(n) < 2n$$

Hence:

$$\boxed{T(n) = O(n)}$$

Also, since the first level itself performs $n$work:

$$\boxed{T(n) = \Theta(n)}$$

Thus, the asymptotic time complexity is **linear**.

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2. Explanation of Work Across Recursive Calls

The work performed at each recursive level is:

Recursion Level	Problem Size	Work
Level 0	n	n
Level 1	$n/2$	$n/2$
Level 2	$n/4$	$n/4$
Level 3	$n/8$	$n/8$
...	...	...
Final	1	1

Total work:

$$n + \frac{n}{2} + \frac{n}{4} + \frac{n}{8} + \cdots + 1$$

This is a geometric progression whose sum is less than $2n$.
Therefore:

$$\boxed{T(n) = \Theta(n)}$$

Although the recursion depth is logarithmic, the total amount of work is dominated by the first few levels, giving **linear time complexity**.

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3. Using Master Theorem

The standard Master Theorem recurrence is:

$$T(n) = aT(n/b) + f(n)$$

Comparing:

$$T(n) = T(n/2) + n$$

We have:

$$a = 1, b = 2, f(n) = n$$

Calculate:

$$n^{\log_b a} = n^{\log_2 1} = n^0 = 1$$

Since:

$$f(n) = n = \Omega(n^{0+\epsilon})$$

for $\epsilon = 1$, Case 3 of the Master Theorem applies.
Therefore:

$$\boxed{T(n) = \Theta(n)}$$

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4. Space Complexity
The recursive problem size is reduced as:

$$n, \frac{n}{2}, \frac{n}{4}, \frac{n}{8}, \dots, 1$$

The recursion depth is:

$$\log_2 n$$

Therefore, if the recursive calls are executed sequentially and the call stack is considered:

$$\boxed{S(n) = O(\log n)}$$

If the algorithm uses additional data structures proportional to the input size, the total space may be $O(n)$.
However, **for the recurrence alone**, the auxiliary space due to recursion is:

$$\boxed{O(\log n)}$$

Final Answer

Aspect	**Complexity**
Recurrence	$T(n) = T(n/2) + n$
Recursion depth	$\log_2 n$
Work per level	$n, n/2, n/4, \dots$
Total work	$n + n/2 + n/4 + \cdots$

Aspect	Complexity
Time Complexity	$\boxed{\Theta(n)}$
Auxiliary Space	$\boxed{O(\log n)}$

Conclusion: The distributed data processing algorithm reduces the problem size by half at every recursive call, but the work performed at each level forms a decreasing geometric series. The total work is therefore $\Theta(n)$, while the recursive call stack requires $O(\log n)$**space**.